

A380X weather radar on the ND and weather on the VD

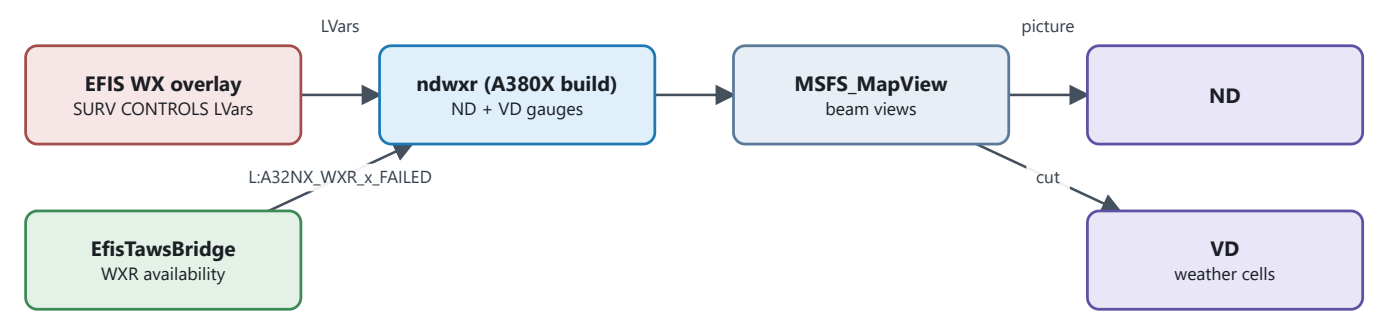
The ndwxr gauge for the A380X, following the EFIS WX overlay and the SURV CONTROLS

Branch	feature/a380x-nd-weather-radar
Aircraft	A380X
Commits	tip 416a0f316 (PR 2), on PR 1
Base	feature/a32nx-nd-weather-radar
Status	In master. Flown. PR draft 02.

Specification

- Availability (EfisTawsBridge): both WXR systems were hard-coded failed (WXR INOP). Each is now failed only with its AESU reset or its power lost (AC ESS system 1, AC 4 system 2).
- ND radar on both A380X NDs, shown with this side's WX overlay and a WXR & TAWS system selected and not failed; range index 0 (OANS) hides it; position IR 1 / 2 or IR 3 per ATT HDG.
- Follows the SURV CONTROLS: WXR, TURB (WX / WX+T; no TURB-only mode on the A380), MODE MAP, WX ON VD.
- VD weather (WX ON VD): the vertical MapView mode is one thin beam, so the VD weather is stylised from the ND radar views along the cut (green / yellow columns, red core, no magenta: FCOM DSC-31-20-40-10).
- 30 s buffer fill after the radar starts transmitting (FCOM operational recommendations).

Component added or changed by the feature MSFS / simulator Aircraft system (systems host, FMS, FWS) Display / page
Data, LVar, storage Crew, flypad, external program



Architecture

The A380's AUTO-mode off-path display was built from a third MapView in the engine's top-view mode and removed: the engine runs one radar per aircraft and that view corrupted the beam views. Both aircraft show the same picture.

PR description (from the PR draft)

From `pr-drafts/02-wx-radar-a380x-vd.md`, section "Summary of Changes" onwards.

Summary of Changes

Adds the weather radar to the A380X NDs with the native `ndwxr` gauge of the A32NX PR, and the weather to the VD.

- **Availability (EfisTawsBridge):** both WXR systems were hard-coded as failed, so the ND always showed WXR INOP as soon as the WX overlay was selected. Each WXR is now failed only while its AESU is reset or its power supply is lost (AC ESS for system 1, AC 4 for system 2), like GPWS.
- **ND radar:** built for the A380X (`build.sh --a380x, ndwxr_A380X.wasm, build-a380x:ndwxr`, igniter task) and added to both ND screens (`vCockpit07` CPT, `vCockpit08` F/O, 768x1024). It follows the A380X's rules: shown when this side's EFIS panel has the WX overlay selected and the WXR & TAWS system selected on the SURV panel is not failed; the ND range table is `[-1, 10 .. 640]` and range index 0 (the OANS map) hides the radar; the ND is on with the DC buses of its display unit (`CdsDisplayUnit` mapping); IR 1/2 feed the CPT/F/O with the ATT HDG knob in NORM, IR 3 otherwise. The ND pages are shared with the A32NX, so the geometry is the same.
- **SURV CONTROLS:** the radar follows the WXR, TURB, MODE (MAP) and WX ON VD buttons (`L:A380X_WXR_OFF`, `_TURB_OFF`, `_MODE_MAP`, `_VD_OFF`, written by the SURV PR). There is no TURB-only mode on the A380X: TURB off gives WX, on gives WX+T. (`PRED WS` and `GAIN` are written by the SURV page too but nothing reads them: MSFS's radar has no predictive windshear or gain.)
- **VD weather (WX ON VD):** the vertical MapView mode of MSFS is a single thin beam (and tilting it crashes the gauge), so the VD weather is stylised from the ND's own radar views: the texel columns near the heading line are rotated onto the range axis and drawn as cells (green and yellow columns with rounded tops, a red core; no magenta: the real VD does not display turbulence, FCOM DSC-31-20-40-10). The heights are fractions of the VD altitude span above the aircraft, placed on the ADR baro-corrected altitude like the VD's own symbol. Single-texel specks and radar sweep streaks are filtered.

Cockpit API Changes

Reads (written by the SURV PR, which documents them): `L:A380X_WXR_OFF`, `L:A380X_WXR_TURB_OFF`, `L:A380X_WXR_MODE_MAP`, `L:A380X_WXR_VD_OFF` (Bool, one per SURV CONTROLS button). Also reads `L:A380X_EFIS_{L,R}_ACTIVE_OVERLAY`, `L:A32NX_WXR_TAWS_SYS_SELECTED`, `L:A32NX_WXR_{1,2}_FAILED`, `L:A32NX_VD_{1,2}_RANGE_LOWER/UPPER` and the ND mode/range LVars. No other LVar added or changed. `panel.cfg` (A380X cockpit part) gets the gauges on the two ND blocks.

Screenshots (if necessary)

TO ADD: before/after of the A380X ND with the WX overlay selected (before: WXR INOP, no weather); CPT with weather and F/O with another range/overlay; the VD with WX ON VD; the SURV CONTROLS page next to the resulting ND.

References

- MSFS SDK, `MSFS_MapView.h` native weather radar API.
- **TO ADD:** A380 FCOM/documentation for the WX overlay of the ND, the WXR & TAWS system selection and the weather on the VD, with screenshots.

Additional context

The VD weather is an approximation (MSFS has no vertical weather data): the shapes are made from the horizontal radar image, they are not real cell heights. Say so in the review if you would rather not ship that part, the ND radar works without it.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. In flight with weather, select the WX overlay on the CPT EFIS panel and a WXR & TAWS system on the SURV panel: weather appears on the CPT ND (ARC and ROSE NAV, ranges 10 to 640), no WXR INOP message. Do the same on the F/O side with another range.
2. SURV CONTROLS page: WXR off removes it, TURB toggles WX / WX+T, MODE MAP removes the precipitation.
3. Range index 0 (OANS) and the PLAN mode hide the radar.
4. WX ON VD: weather cells appear on the VD around the aircraft altitude, and disappear with WX ON VD off.
5. Reset the AESU of a WXR system, or remove AC ESS / AC 4: that system is failed (WXR INOP) and its weather goes.
6. ATT HDG knob away from NORM: the position source changes, no glitch.